



ROWAD
MODERN ENGINEERING

Risk Assessment Process
P-QD-05

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	Risk Assessment Process		
	Effective Date: Nov, 2021	Rev. No.: 01	Doc. No.: P-QD-05

1. OBJECTIVES

The objective of this process is to ensure that:

- A systematic and holistic approach to identify risks
- Analyse the likelihood of occurrence and impacts of these risks, in order to achieve the project objectives in terms of time, cost, quality, safety and environmental aspects.

2. STAKHOLDER

- RME departments' Managers
- PMs

3. LIABLE STAKEHOLDER

- RME departments' Managers
- PMs

4. TERMS AND DEFINITION

- **PM:** Project Manager
- **QA:** Quality Assurance
- **HoD:** Head of Department
- **Risk:** Effect of uncertainty on objectives. An effect is a deviation from the expected positive or negative. Risk is about what could happen & what the effect of this happening might be.
- **Opportunity:** Opportunities can arise as a result of a situation favorable to achieving an intended result
- **Risk Management:** Coordinated activities to direct and control an organization with regards to risk
- **Risk assessment:** Overall process of risk identification, risk analysis and risk evaluation
- **Consequence:** Outcome of an event affecting objectives
- **Likelihood:** Chance of something happening

5. RELATED DOCUMENTED INFORMATION

- Risk Register form, F-P-QD-05-01
- Action plan to address risks and opportunities, F-P-QD-05-02

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6. PROCESS DETAILS

6.1. PROCESS INPUT

- Departments processes
- RME Projects information/contract/condition

6.2.PROCESS OPERATION

Sr.	Process Cycle description	Responsibility	Form
1	<u>Development of risk register- Pre award stage</u> Commercial dept shall prepare the risk assessment report within 7 working days from receiving the client's documents and send it to the Tender department. Commercial dept shall identify for the potential awarded projects. Accordingly, Tender dept shall trigger the release of the risk register to the following department Managers: • Operation • HSE • Quality Control • Projects Control • MEP • Purchasing • Procurement Department Managers shall identify the risks, determining the risk grade by combining Likelihood and Seriousness. Tender dept shall estimate the contingency cost accordingly. <u>Development of risk register- Post award stage</u> Department Manager/PM shall prepare the risk assessment every two quarters by using the risk	Commercial Tender Department Manager PM	Risk Register F-P-QD-05-01

Sr.	Process Cycle description	Responsibility	Form										
	register form as shown in the following points, then send it to the Quality Assurance department. Development of the risk register shall take place using the steps below												
1.1	A. Identify the sources of risks related to the processes and the associated uncertainties, considering relevant triggers that may cause the risk to be realized												
1.2	B. Describe the nature of the risk, and the impact on the project/department if the risk is not mitigated or managed.												
1.3	C. Identify the Likelihood for each Risk according to the following criteria <table><tr><th>Likelihood</th><th>Meaning</th></tr><tr><td>Low</td><td>Very unlikely to occur</td></tr><tr><td>Medium</td><td>Likely to occur sometimes</td></tr><tr><td>High</td><td>High Likely to occur</td></tr></table>	Likelihood	Meaning	Low	Very unlikely to occur	Medium	Likely to occur sometimes	High	High Likely to occur				
Likelihood	Meaning												
Low	Very unlikely to occur												
Medium	Likely to occur sometimes												
High	High Likely to occur												
1.4	D. Identify the Seriousness for each Risk according to the following criteria <table><tr><th>Seriousness</th><th>Meaning</th></tr><tr><td>Low</td><td>Has low impact</td></tr><tr><td>Medium</td><td>Impact sometimes</td></tr><tr><td>High</td><td>High impact</td></tr><tr><td>Extreme</td><td>Extreme Impact</td></tr></table>	Seriousness	Meaning	Low	Has low impact	Medium	Impact sometimes	High	High impact	Extreme	Extreme Impact		
Seriousness	Meaning												
Low	Has low impact												
Medium	Impact sometimes												
High	High impact												
Extreme	Extreme Impact												

Sr.	Process Cycle description	Responsibility	Form																														
1.5	E. Determine the risk grade by combining Likelihood and Seriousness, which represents the risk priority, criticality and importance as following: <table><tr><th><u>Likelihood</u></th><th><u>Seriousness</u></th><th><u>Grade</u></th></tr><tr><td rowspan="4">Low</td><td>Low</td><td>D</td></tr><tr><td>Medium</td><td>D</td></tr><tr><td>High</td><td>C</td></tr><tr><td>Extreme</td><td>A</td></tr><tr><td rowspan="4">Medium</td><td>Low</td><td>D</td></tr><tr><td>Medium</td><td>C</td></tr><tr><td>High</td><td>B</td></tr><tr><td>Extreme</td><td>A</td></tr><tr><td rowspan="4">High</td><td>Low</td><td>C</td></tr><tr><td>Medium</td><td>B</td></tr><tr><td>High</td><td>A</td></tr><tr><td>Extreme</td><td>A</td></tr></table> Grade A: Mitigation actions, to reduce the likelihood and seriousness, to be identified and implemented as soon as the project commence as a priority. Grade B: Mitigation actions, to reduce the likelihood and seriousness, to be identified and appropriate actions implemented during project execution. Grade C: Mitigation actions, to reduce the likelihood and seriousness, to be identified and costed for possible action if funds permit. Grade D: To be noted – no action is needed unless grading increases over time.	<u>Likelihood</u>	<u>Seriousness</u>	<u>Grade</u>	Low	Low	D	Medium	D	High	C	Extreme	A	Medium	Low	D	Medium	C	High	B	Extreme	A	High	Low	C	Medium	B	High	A	Extreme	A		
<u>Likelihood</u>	<u>Seriousness</u>	<u>Grade</u>																															
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	High	A																															
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Sr.	Process Cycle description	Responsibility	Form
1.6	F. Specify planned mitigation strategies according to the following cases: • Preventative (implement immediately) • Contingency (implement if/when risk occurs)		
1.7	G. Specify who is responsible for undertaking each mitigation action(s)		
1.8	H. Specify the expected cost for taking the mitigation action		
1.9	I. Specify the timeframe for mitigation action(s) to be completed		
1.10	J. Specify the change in Grade and the date since last review, if any.		
2	Department Manager shall plan actions to address risks and Opportunities by using action plan form which includes the following: • Description of the actions • Responsibilities • Due dates • Resources • Evaluation criteria (that evaluate the effectiveness of these actions) • Follow up	Department Manager	Action plan to address risks and opportunities F-P-QD-05-02
3	Departments' mangers shall Identify opportunities	Department Manager	Action plan to address risks and opportunities

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	that can arise from the risks to: • Attract new customers • Opening new market • Building partnerships • Adoption of new practices • Develop new products and services • Improve productivity • Using new technology		F-P-QD-05-02
4	<u>Description of Risk management</u> I. Risk Management Planning Risk management planning is the systematic process of deciding how to approach, plan, and execute risk management activities throughout the life of a project/process. It is intended to maximize the beneficial outcome of the opportunities and minimize or eliminate the consequences of adverse risk events		
5	II. Identify Risk Events Risk identification involves determining which risks might affect the project/process and documenting their characteristics. It may be a simple risk assessment or an outcome of a workshop process.		
6	III. Qualitative Risk Analysis Qualitative risk analysis assesses the impact and likelihood of the identified risks, and develops prioritized lists of these risks for further analysis or direct mitigation. Departments' managers assess		

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	each identified risk for its probability of occurrence and its impact on project/process objectives.		
7	IV. Risk Response Risk response is the process of developing options and determining actions to enhance opportunities and reduce threats to the process/project's objectives. It identifies and assigns parties to take responsibility for each risk response. This process ensures each risk requiring a response has an "owner." The Departments' managers identify which strategy is best for each risk, and then select specific actions to implement that strategy.		
8	V. Opportunities Check the suitability, adequacy and effectiveness of the Quality Management System based on the results of analysis and evaluation with the outputs of management review. Actions taken to address opportunities shall be proportionate to the proportional impact on the conformity of products and services.		
9	VI. Risk Monitoring and Control Risk monitoring and control tracks identified risks, monitors residual risks, and identifies new risks ensuring the execution of risk plans and evaluating their effectiveness in reducing risk.		

6.3. PROCESS OUTPUT

- Risk identification and assessment Corrective action

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7. PROCESS RISK ASSESSMENT

Included in the risk register form No. F-P-QD-05-01 "Risk identification and assessment" maintained by the HoD, along with the action plan to address risks and opportunities".

8. PROCESS CONTROL.

- The Departments' managers shall monitor the risks every two quarters, and in the following cases:
 - When changes occur in internal or external Issues
 - When action is taken will change the context and the risk
 - When changes occur in the Requirements of Interested Parties
 - When Changes Occurs in the processes
- The effectiveness of actions taken to address risks and opportunities shall have reviewed at least once a year during the management review meeting conducted by QA manger, and their records shall be updated accordingly

9. PERFORMANCE MEASURE

- The change in grade for each risk with every update.

10.— DOCUMENT CHANGE HISTORY

No.	Revision Level	Description of changes	Effective Date
1	00	Major Changes due to transition from ISO 9001:2008 to 9001:2015	1/12/2016
2	01	All points	Nov, 2021